Anjali Katasani
Sr Azure Data Engineer
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PROFESSIONAL SUMMARY

- Over 10+ years of experience as a Senior Data Engineer, specializing in data architecture, ETL development, data pipeline optimization, and cloud-based data solutions.
- Expertise in Azure Data Factory (ADF) and Snowflake, with a proven track record of designing, developing, and optimizing ETL pipelines and automated workflows.
- Strong background in data quality monitoring, troubleshooting ETL pipeline failures, and optimizing data performance for large-scale data processing.
- Deep understanding of Medicaid, Medicare, and healthcare data processing, including expertise in HL7 and FHIR standards for structured healthcare data management.
- Hands-on experience in building APIs in Snowflake and ADF (REST, RPC) to facilitate seamless data exchange and system interoperability.
- Expertise in processing structured and unstructured data using Azure Data Services, enhancing data integration capabilities.
- Optimized Databricks clusters with auto-scaling and cost-efficient configurations, improving performance while reducing cloud expenses.
- I am skilled in GitHub, SVN, and CI/CD pipelines, ensuring efficient version control, deployment automation, and integration across cloud and on-premise environments.
- Proven ability in performance tuning and database optimization, accelerating data retrieval speeds, and improving overall system efficiency.
- Implemented Azure Key Vault for secure credentials management, ensuring compliance with enterprise data security standards.
- Designed Snowflake Task Scheduling workflows to automate data processing and manage dependencies for seamless execution.
- Implemented Databricks Delta Live Tables (DLT) for real-time data processing and quality monitoring
- Developed custom shell scripts for data loading to HDFS, ensuring efficient data transfer from various sources and optimizing edge node workflows in big data environments.
- Worked with Oozie and Airflow to orchestrate and automate data pipelines, enabling reliable workflow scheduling and management of data processing tasks in production.
- Deployed Databricks in a multi-cloud environment, ensuring scalability and cost optimization across platforms
- Led SQL database migration projects to Azure Data Lake, Data Lake Analytics, and Azure SQL, handling data movement and security and granting database access for seamless cloud integration.
- Skilled in Amazon Web Services (AWS), including EC2, S3, VPC, IAM, and Redshift, among others, managing cloud environments to provide scalable, cost-effective data solutions and infrastructure management.
- Created custom UDFs in Python to extend Hive and Pig functionality, adding unique capabilities to big data processing and enhancing data manipulation within these ecosystems.
- Implemented vector search on Databricks using optimized embeddings for fast retrieval
- Optimized data export and loading with Teradata utilities like Fast Export and Multiload, successfully managing data integration across flat files and diverse source systems.
- Built automated regression scripts in Python to validate ETL processes across multiple databases (Oracle, SQL Server, Hive, MongoDB), ensuring data integrity and quality at every stage.
- Managed data import and ETL processes from Oracle DB to S3 and Redshift using Big Data Manager, automating data flows for increased efficiency and reliability in data warehousing.
- Built lineage tracking in Databricks, enhancing data observability and traceability across pipelines.

- Designed complex data warehouse schemas, including star and snowflake schemas, applying best practices in SQL, MySQL, PostgreSQL, and Redshift for effective data storage and access in large-scale systems.

AREAS OF EXPERTISE

Big Data Technologies:	Hadoop, Spark Core, Spark SQL, Hive, HBase, Parquet, Sqoop, Pig, Apache Kafka, Zookeeper, Flume, MapReduce Design Patterns, Data frames, Spark Streaming, Python, Hadoop Distribution (Cloudera, Horton Works), PySpark, Scala.
Data Integration & ETL:	SQL, Azure Data Factory, Snowflake Modeling, Data Flow, Apache Kafka, Apache Flume, Apache Nifi, Apache Spark, Apache Beam, Apache Airflow, ELT/ETL Pipelines with Python, Snowflake Snow SQL, Spark Data Frames, Logic Apps, Azure Functions, CI/CD Frameworks (Jenkins), Data Ingestion, Azure Synapse Analytics, Vertica, Polybase
Programming Languages:	SQL, PL/SQL, Python, Scala.
Version Control:	GIT, GitHub.
Databases:	Azure SQL DB, Azure Synapse, Oracle, Snowflake, PostgreSQL, Cosmos DB
Data Storage & Retrieval:	Azure Storage, Azure SQL DB, Blob Storage, MySQL, Cassandra, Snowflake, Azure Data Lake Storage (ADLS), Gen2, Azure Data Bricks, Data Lakes, HDFS
Data Analysis & Reporting:	Power BI, Tableau
Collaboration & Project Management:	Agile Methodology, Jenkins, Terraform, Azure DevOps, GitHub Actions.

EDUCATION:

- Bachelor's in computer science from GITAM Deemed University

CERTIFICATIONS:

- Microsoft Certified: Azure Data Engineer – Associate
- Databricks Academy Accreditation - Platform Administrator

WORK EXPERIENCE:

Client: QBE North America, NY **June 2023 to Present**
Role: Sr. Data Engineer

- Engineered and implemented an automated feed file generation system for Azure Blob Storage, optimizing data organization and storage efficiency for critical information.
- Managed end-to-end ETL operations using Azure Data Factory, showcasing proficiency in orchestrating the extraction, transformation, and loading of data at large scale.
- Built scalable ETL/ELT pipelines in Databricks, enabling efficient data ingestion, transformation, and analytics for large-scale insurance and retail datasets.
- Leveraged Azure Synapse Analytics for large-scale data processing, demonstrating competence in handling complex analytical workloads.
- Optimized Databricks workflows for processing high-volume claims, customer transactions, and operational data, ensuring real-time insights.
- Organized end-to-end ETL pipelines in Azure, seamlessly integrating Azure Databricks for large-scale big data processing, demonstrating a holistic and efficient approach to data processing and transformation.
- Pioneered real-time data processing using Azure Stream Analytics, highlighting agility in managing and analyzing data streams for timely insights and decision-making.
- Designed Databricks-based data lakehouse architecture, unifying raw, curated, and transformed data for advanced reporting
- Spearheaded the containerization of data processing apps using Azure Kubernetes, emphasizing commitment to scalable and portable solutions.

- Implemented Databricks Delta Lake for ACID-compliant data lakes, improving reliability, consistency, and versioning.
- Implemented Apache Spark for distributed data processing, harnessing its parallel computing capabilities to efficiently handle large-scale data analytics and processing tasks.
- Implemented anomaly detection in Databricks to identify fraudulent transactions, policy irregularities, and inventory shrinkage patterns.
- Leveraged Power BI for intuitive data visualization and interactive business intelligence reporting, providing actionable insights and facilitating informed decision-making within the organization.
- Developed and optimized SQL queries for efficient database operations, ensuring reliable data retrieval, manipulation, and management in support of business-critical applications.
- Implemented Slowly Changing Dimensions (SCD) techniques in data warehousing, enabling the effective tracking and management of historical changes in data, crucial for maintaining accurate and meaningful analytics over time.
- Designed and implemented data warehouse architecture using Snowflake, leveraging its cloud-native platform to achieve scalable, elastic, and high-performance analytics with a focus on simplicity and ease of maintenance.
- Organized real-time data streaming and event-driven architectures by implementing Apache Kafka, ensuring reliable, scalable, and fault-tolerant data pipelines for seamless communication and processing across distributed systems.
- Established and automated complex workflows by deploying Apache Airflow, enhancing data pipeline orchestration, scheduling, and monitoring for improved efficiency and reliability in data processing tasks.
- Implemented a data build tool to automate the end-to-end process of extracting, transforming, and loading (ETL) data, streamlining data workflows, and ensuring consistent, reliable data outputs for analytics and reporting.
- Developed Data Definition Language (DDL) scripts for Stage and Operational Data Store (ODS) tables in Azure SQL Database, demonstrating expertise in database design and meticulous management for efficient data storage and retrieval.
- Led the development of DBT models, macros, and data marts in Snowflake or Azure Synapse Analytics, showcasing proficiency in crafting efficient and structured data models to support advanced analytics and reporting needs.
- Innovated by crafting custom activities using Azure Functions and PowerShell scripts, streamlining processes, and significantly enhancing operational efficiency through automation and tailored solutions.
- Implemented automation using Azure Logic Apps, with integration into Azure Key Vault for enhanced security measures and data governance.
- Implemented Apache Airflow in Azure for HiveQL scripts and jobs, orchestrating complex workflows for seamless execution.
- Spearheaded the implementation of CI/CD pipelines and frameworks with Azure DevOps, ensuring a streamlined and automated development lifecycle.
- Actively participated in Agile ceremonies and collaborated with DevOps engineers, utilizing JIRA and Azure DevOps for effective project management and collaboration.
- Documented technical specifications, data flow diagrams, and process documentation for Azure Architecture, ensuring clear and comprehensive documentation for knowledge transfer and future reference.

Environment: Azure SQL Database, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Kubernetes, Azure DevOps, ADLS Gen2, Event Hubs, PolyBase, PySpark, HDFS, MapReduce, Spark, Spark Streaming, Kafka, Hive, ORC, Parquet, Text File, Compression Techniques, Dimensional Modeling, Star Schema, Snowflake Schema, SQL Server, Oracle, Vertica, Teradata.

Client: Meijer, Grand Rapids, MI **Aug 2020 to May 2023**
Role: Azure Data Engineer

- Implemented efficient data integration solutions to seamlessly ingest and integrate data from diverse sources, including databases, APIs, and file systems, using tools like Apache Kafka, Apache NiFi, and Azure Data Factory.
- Enhanced data fetching efficiency by 40% through the implementation of optimized query techniques and indexing strategies, ensuring scalability and smooth functioning of ETL data pipelines.
- Data Ingestion to various Azure Services like Azure Data Lake, Azure Storage, Azure SQL, Azure DW, and processing the data in Azure Databricks.
- Worked on Microsoft Azure services like HDInsight Clusters, BLOB, Data Factory, and Logic Apps and did Proof of Concept (POC) on Azure Data Bricks.
- Perform ETL using Azure Data Bricks, Migrated on-premises Oracle ETL process to Azure Synapse Analytics.
- Worked on Migrating SQL database to Azure Data Lake storage (adls), Azure Data Lake analytics, Azure SQL Database, Data Bricks, and Azure SQL Data Warehouse.
- Expertise in leveraging Azure Synapse and PolyBase for efficient data transfer, seamless integration between disparate data sources and data lakes, and facilitating streamlined ETL processes and real-time analytics for Azure-based data engineering solutions.
- Deployed and optimized Python web applications to Azure DevOps CI/CD to focus on development.
- Developed enterprise-level solutions using batch processing and streaming framework using Spark Streaming and Apache Kafka.
- Designed and implemented robust data models and schemas to support efficient data storage, retrieval, and analysis using technologies like Apache Hive, Apache Parquet, and Snowflake.
- Developed and maintained end-to-end data pipelines using Apache Spark, Apache Airflow, or Azure Data Factory, ensuring reliable and timely data processing and delivery.
- Provided production support and troubleshooting for data pipelines, identifying and resolving performance bottlenecks, data quality issues, and system failures.
- Created Partitions and Buckets based on State to further process using Bucket-based Hive joins.
- Worked with Data Lakes and big data ecosystems (Hadoop, Spark, Hortonworks, Cloudera).
- Load and transform large sets of structured, semi-structured, and unstructured data.
- Written Hive queries for data analysis to meet the specified business requirements by creating Hive tables and working on them using Hive QL to simulate MapReduce functionalities.
- Proficient in leveraging PostgreSQL for efficient data storage, management, and querying, ensuring optimal performance and reliability in data pipelines and analytical workflows.
- Proficiency in optimizing performance and ensuring reliability of Azure data solutions through proficient utilization of .NET framework tools and best practices.
- Skilled in implementing event-driven architectures using Azure Event Hubs, enabling real-time ingestion, processing, and analysis of streaming data at scale.
- Proficient in designing and managing event queues with Azure Service Bus, facilitating reliable and asynchronous communication between decoupled components in distributed systems.
- Expertise in administering Teradata environments, including installation, configuration, monitoring, and maintenance tasks, ensuring high availability, reliability, and scalability of the data infrastructure to meet business requirements.
- Proficiency in configuring linked services, utilizing Copy Activity for data movement, and Lookup Activity for data enrichment and validation within Azure Data Factory.
- Used Spark Streaming to divide streaming data into batches as input to Spark engine for batch processing.
- Implemented CI/CD pipelines that resulted in a 50% reduction in deployment time and a 30% decrease in the occurrence of deployment-related errors in the Hadoop environment.
- Used JIRA to manage the issues/project workflow.

- Utilized Git and GitHub repositories to maintain the source code and enable version control.

Environment: Azure Databricks, Azure Data Factory, Azure Logic Apps, Functional App, HDFS, MapReduce, Apache Spark, Apache Hive, Python, Scala, PySpark, PostgreSQL, .NET, GIT, JIRA, Jenkins, Apache Kafka, Power BI, Azure Synapse Analytics, Polybase, Azure Data Lake Storage, Azure SQL Database, Azure DevOps.

Client: Micron Technology, Boise, ID. **Oct 2017 to July 2020**
Role: Data Engineer

- Design & implement migration strategies with Azure suite: Azure SQL Database, Azure Data Factory (ADF) V2, Azure Key Vault, and Azure Blob Storage.
- Created end-to-end Data pipelines using ADF services to load data from On-Prem to the Azure SQL server for Data orchestration.
- Build scalable and reliable ETL pipelines to pull large and complex data from different systems efficiently.
- Worked on building the data pipeline using Azure Services like Data Factory to load the data from Legacy, SQL server to Azure Data warehouse using Data Factories and Databricks Notebooks.
- Develop Spark applications using PySpark and Spark SQL for data extraction, transformation, and aggregation from multiple file formats for analyzing and transforming.
- Experience with RESTful API development, including defining resources, handling requests, and returning responses in JSON or XML formats.
- Expertise you have with building Kafka-based data pipelines, including data ingestion, processing, and analysis.
- Experience with Kinesis APIs, including Kinesis Client Library and Kinesis Producer Library.
- Proficiency with Python libraries commonly used in data engineering, such as Pandas, NumPy, PySpark, and Scikit-learn.
- Expertise using Pub/Sub messaging systems in combination with big data technologies, such as Hadoop, Spark, or Flink.
- Experience you have with building CDC pipelines, including designing schema changes, capturing changes in real-time, and applying changes to downstream systems.
- Developed Python-based API (RESTful Web Service) using Flask. Involved in the Analysis, Design, and Development, and Production phases of the application.
- Developed Python batch processors to consume and produce various feeds.
- Constructed product-usage data aggregations using Py-Spark and Spark SQL and maintained in Azure Data warehouse for reporting, data science dashboarding, and ad-hoc analyses.
- Leveraged Jenkins for CI/CD Deployments and releases using Git and GitHub Integrations.
- Utilized Agile process and JIRA issue management to track sprint cycles.

Environment: Azure SQL Database, Azure Data Factory (ADF) V2, Azure Key Vault, Azure Blob Storage, PySpark, Spark SQL, RESTful API, Kafka, Kinesis APIs, Python libraries (Pandas, NumPy, Scikit-learn), Pub/Sub messaging systems, Hadoop, Spark, Flink, CDC pipelines, Flask, Jenkins, AJAX, JSON, Git, GitHub, Agile, JIRA.

Client: Franklin Templeton Investments, India **July 2014 to Dec 2016**
Role: Software Engineer

Responsibilities

- Proficient in developing stored procedures, optimized triggers, essential functions, and implementing performance-enhancing indexes and indexed views, contributing to improved query performance and overall efficiency.
- Expertise in effectively monitoring and optimizing SQL Server Performance, employing industry best practices and techniques to ensure efficient and high-performing database systems.
- Experienced in designing ETL data flows using SSIS, developing mappings and workflows for seamless data extraction from SQL Server, and performing efficient data migration and transformation from Excel sheets using SQL Server SSIS.
- Extensively wrote the PL/SQL and SQL programs and designed and developed the views, materialized views, functions, procedures, packages, triggers, and cursors.
- Expertise in finding Facts and Dimensions, creating fact and dimension tables, and managing Slowly Changing Dimensions (SCD) for efficient dimensional data modeling for Data Mart design.
- Proficient in error and event handling techniques, including the utilization of precedence constraints, breakpoints, checkpoints, and logging mechanisms, to ensure comprehensive error detection, resolution, and effective event management.
- Expertise in creating business Building Cubes and Dimensions, incorporating various architectural designs and data sources, and utilizing MDX scripting to enable advanced analytics and reporting functionalities.
- Extensive understanding of the features, structure, attributes, hierarchies, and schema designs such as Star and Snowflake Schemas in Data Marts, demonstrating proficiency.
- Proficient in utilizing Python for data manipulation, analysis, and automation tasks. experienced in utilizing ERWIN as a data modeling tool to design and manage data models in data warehouse development.
- Skilled in SSAS cube development, encompassing key areas such as aggregation, KPIs, measures, cube partitioning, data mining models, and deployment and processing of SSAS objects.
- Created diverse reports (list, dashboard, cross-tab, and charts) using Tabular SQL, Tabular S, and Tabular Sets, resulting in improved report performance and reduced load on the Framework Manager model.
- Experienced in leveraging NoSQL databases MongoDB and Cassandra to handle and analyze large volumes of unstructured and semi-structured data in data warehouse environments, enabling scalable and flexible data storage and retrieval.
- Proficient in data visualization and analysis using Tableau, creating interactive dashboards and reports to effectively communicate insights and support data-driven decision-making.
- Extensively worked on Performance Tuning and Query Optimization using various types of Hints, Partitioning, Bulking Techniques, and indexes.
- Skilled in SharePoint administration, adept at implementing and optimizing SharePoint solutions to facilitate document management, team collaboration, and business process automation for enhanced efficiency and productivity.
- Experience in managing complex data models with Erwin's Model Mart feature.
- Demonstrated extensive experience in utilizing Unix Shell Scripts for efficient support and server-based tasks.
- Expertise in developing Parameterized charts, graphs, Linked, Dashboard, Scorecards, Report on SSAS Cube using Drill-down, Drill-through and Cascading reports using SSRS.
- Proficient in utilizing GIT as a version control tool for effective code repository maintenance, facilitating superior code management.

Environment: MS SQL Server 2016, Visual Studio 2017/2019, SSIS, Share point, MS Access, Team Foundation server, Git.